higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

DATA COMMUNICATION AND NETWORKING
NQF LEVEL 4

18 NOVEMBER 2019

This marking guideline consists of 8 pages.

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SECTION A

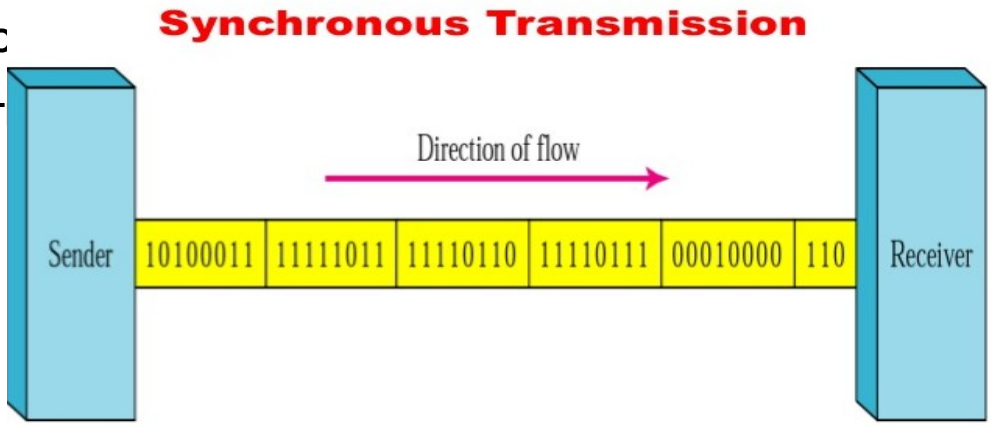
QUESTION 1

1.1	1.1.1	D		
	1.1.2	C		
	1.1.3	A		
	1.1.4	A		
	1.1.5	C		
	1.1.6	D		
	1.1.7	C		
	1.1.8	C		
	1.1.9	A		
	1.1.10	B		
			(10 × 1)	(10)
1.2	1.2.1	Integrated		
	1.2.2	DSL		
	1.2.3	Serial		
	1.2.4	MMF/multimode fibre		
	1.2.5	Scheduled tasks		
	1.2.6	User requirements		
	1.2.7	Network		
	1.2.8	Suite		
	1.2.9	Tune		
	1.2.10	Data stream transmission		
			(10 × 1)	(10)
1.3	1.3.1	False		
	1.3.2	True		
	1.3.3	True		
	1.3.4	False		
	1.3.5	False		
	1.3.6	True		
	1.3.7	True		
	1.3.8	True		
	1.3.9	False		
	1.3.10	True		
			(10 × 1)	(10)
1.4	1.4.1	F		
	1.4.2	M		
	1.4.3	A		
	1.4.4	L		
	1.4.5	C		
	1.4.6	E		
	1.4.7	D		
	1.4.8	J		
	1.4.9	B		
	1.4.10	I		
			(10 × 1)	(10)
			TOTAL SECTION A:	**40**

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SECTIC

QUEST



2.1 A – wireless router✓
B – modem or wireless access point✓ (2)

2.2 Wireless router
- used to provide access to the internet✓
- used for routing✓
- as a wireless access point to provide connectivity for wireless devices.✓ (ACCEPT ANY 2)

Modem
- converts digital signals into analog signals and vice versa✓
- used to connect to the internet✓

Wireless access point
- connects a group of wireless devices, that is a wireless network to a wired LAN✓
- connects wireless devices and allows them to send messages to each other✓ (4)

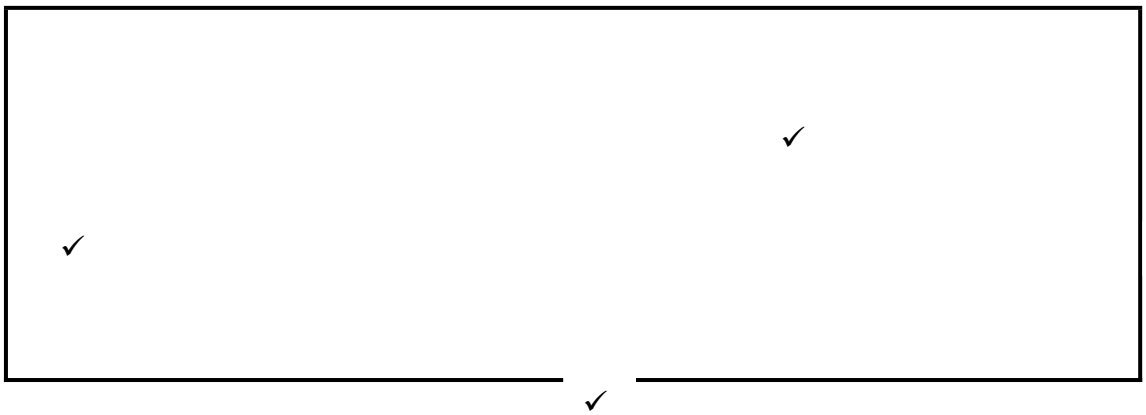
2.3 Option B is used in a large network with many PCs to configure✓and when configuring end users' PCs✓ (2)

2.4 Dynamic host configuration protocol or DHCP✓ (1)

2.5 192.168.1.51 (NO alternate answer to be accepted.) (2)

2.6 A – data flow
B – sender
C – start bits
D – stop bits (4)

2.7



(4)
[19]

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QUESTION 3

3.1
- Framing✓
  Data link layer protocols encapsulate or package massages into frames that is, creates frames.✓
- Physical addressing✓
  Data link layer protocols add a unique address (Physical address or MAC address) to the frame to identify the sender and receiver. ✓
- Flow control✓
  Data link layer protocols controls the flow of data to ensure that the destination does not receive more data than it can handle.✓
- Error control✓
  Data link layer protocols detect and retransmit damaged or lost frames or messages.✓
- Access control✓
  When two or more devices are connected to the same network or link, data link layer protocols are necessary to determine which device has access to the network or link at any given time.✓ (4 × 2) (8)

3.2 3.2.1 Internet of Things or IoT
The interconnection via the Internet of computing devices embedded in everyday objects✓ such as cars, homes etc. These objects are installed with chips, sensors and software enabling them to connect to the Internet.✓

3.2.2 Cloud computing
The practice of using a network of remote servers hosted on the internet✓ to store, manage and process data, rather than a local server or a personal computer.✓

3.2.3 Smart personal assistants
It is a mobile software that can perform tasks or services on behalf of an individual✓ based on a combination of user inputs, location awareness and the ability to access information from a variety of online sources (such as weather conditions, traffic congestion, news, stock prices, user schedules, retail prices, et cetera).✓

3.2.4 Wearable technology
These are smart electronic devices that can be incorporated into clothing or worn on the body as implants or accessories.✓ Wearable devices such as activity trackers enable objects to exchange data through the internet with a manufacturer, operator and/or other connected devices, without requiring human intervention.✓

(4 × 2) (8)
[16]

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QUESTION 4

4.1
- Fires
- Flooding
- Earthquakes
- Lightning strikes
- Vandalism
- Power surge
- Hacker attack
- Misconfiguration/ configuration errors
- Virus attack
- Theft
- Equipment failure
- Loss of trained IT staff (Any 6 × 1) (6)

4.2 4.2.1 No. ✓ A switch does not need an IP address for it to work that is, to do switching. ✓ (ONE mark if only NO is answered) (2)

4.2.2 Mac-address table or content-addressable memory (CAM) table.✓ (1)

4.3 Give the users at least a few days' notice for planned downtime.✓ This gives them enough time to prepare themselves. (1)

Installations should be scheduled for times such as:

- Quiet periods (for example, during holiday periods)
- Before or after normal office hours
- While the client/user is out of the office
- When the client will not need his/her computer

(1)
(ONE for these four points)

- Let users know what new features or improvements this upgrade will bring.
- Explain what parts of the infrastructure are affected, what services are affected and what is not affected.
- Give the exact start time and estimated end time of the installation.
- If the users need to take action, include very clear instructions on what to do.
- Give a contact name/number to provide assistance.
- Assure users that the situation is under control.
- Include contact information or methods to receive additional information or updates. (7)

[18]

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QUESTION 5

5.1
- Mapping network drives
- Installing and setting a user's default printer
- Collecting computer system information
- Updating virus signatures
- Updating software (5)

5.2 You will have to open up the network and sharing centre.✓ Click the Ethernet link at the top of the window, open up the properties box. Click Internet Protocol Version 4(TCP/IPV4) and open the properties box.✓ Select the option 'use the following IP address'. Type an address into the IP address field. Type an address into the Subnet mask field.✓ Type an address into the Default gateway field. Click OK. This will save the settings for that computer.✓ (4)

5.3
- Open the *Command Prompt* on the computer, type *shutdown/i* and press *enter*. This will start the *Remote Shutdown utility* in a separate window.✓
- Click the *Add*. This will let you add the computer on your network for which you want to manage the shutdown process.✓
- Enter the *computer's name* and then click *'OK'* to add it to the list.✓
- Set your *shutdown options*. There are several options you can set before you send the *shutdown signal*:
  - You can choose to shut down the remote computer or restart it.✓
  - You can warn the users that their computers will be shut down. This is highly recommended if you know people are using the computers. You can adjust the length of the warning that is displayed.✓
  - You can add a reason and comment at the bottom of the Window. These will be added to the logs, which is important if you have multiple administrators or need to be able to review your actions later.✓
- Click *'OK'* to shut down the remote computers.✓
- If you set a warning time, the computers will shut down when it expires, otherwise they will shut down immediately✓ (8)

[17]

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QUESTION 6

VPNs (virtual private networks), which provide secure connections between individual users and their organisation's network over the internet, have several benefits.✓ They provide more secure site-to-site connections and they transfer information much faster than WANs.✓ VPNs are much less expensive,✓ as you can use a single leased line to the internet for each office, cutting down on broadband costs. VPNs can be setup in the shortest possible time as compared to normal WANs where one usually has to apply for a leased line and this takes time.✓

While VPNs are highly recommended and trusted, a new technology gaining traction with companies is the remote desktop. Remote desktops require software or an operating system that allows applications to run remotely on a server,✓ but to be displayed locally simultaneously.✓ This network server hosts remote files and applications in a secure location, ensuring your data is never lost.✓ Remote desktops help you control where the data is located✓ so that you don't have data roaming out there in the world.✓

Cloud computing has several advantages and functions that simplify IT management for the company.✓ It reduces costs as a company will not need to invest in servers and server applications as these can be leased from cloud computing providers. ✓ It allows you to set up what is essentially a virtual office to give you the flexibility of connecting to your business anywhere, any time.✓ It has the ability to communicate and share more easily outside the traditional methods.✓ It reduces your operating costs as both server software and user applications do not need to be installed on local computers therefore no license fees as well IT support staff fees required. ✓ (Any 10 × 1) **[10]**

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QUESTION 7

7.1
- Data source/ sender
- Protocols
- Message
- Medium
- Destination/ receiver (5 × 1) (5)

7.2 Local area networks are computer networks that cover small geographic areas✓ usually small buildings or a group of buildings✓whereas wide area networks can cover large geographic areas✓such as countries or provinces.✓ (4)

7.3 7.3.1 Logic link control sublayer✓ and MAC sublayer✓ (2)

7.3.2 The logic link control sub layer is responsible for control. It controls the flow of data.✓ It ensures that the destination does not receive too much data.✓ It performs error control by detecting and retransmitting dropped packets.✓

The MAC sublayer makes sure that each device on a network has a unique address in order to receive data.✓ This unique address given to the network device is called a MAC address.✓ It is hard-wired into the device before it leaves the manufacturer.✓ (6)

7.4 Broadband refers to a network that can handle a much broader range of frequencies✓ This means the broader the range the higher the bandwidth✓and it also used to refer to high speed internet.✓ (3)

[20]

QUESTION 8

8.1
- Keep users informed of what is happening.
- Network must be laid out on paper.
- Obtain diagrams of existing cabling.
- Perform site inspection.
- Test components.
- Calculate cable lengths.
- Record where in the network each component will be connected.
- Burn in all hardware components.
- Test entire system and also test software.
- Train users. (Any 8 × 1) (8)

8.2
- Higher bandwidth or speed than copper
- Immune to electromagnetic interference (2)

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- Covers a greater distance than copper (Any 2 × 1)

[10]

TOTAL SECTION B: 110
GRAND TOTAL: 150

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